

SPRING
1946

PLANET STORIES 20c

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HATH ME?**
An Experiment in
Thrallmery....by
HENRY KUTTNER

TWO BIG COMPLETE NOVELS
The **Blue Venus** || **Engines of the Gods**
by EMMETT McDOWELL || by GARDNER F. FOX

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*** START OF THE PROJECT GUTENBERG EBOOK ELECTRON EAT
ELECTRON ***

Electron Eat Electron

By NOEL LOOMIS

*(Editor's note: When we had read through
this in-a-class-by-itself story, we exclaimed,
"Here's PLANET'S scoop on the world!" What do
you think? Does Mr. Loomis answer the
questions: "How will future wars be fought?
Will civilization be destroyed?")*

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Supreme General Hoshawk, chief of staff, watched with piercing gray eyes while the President of the United States of the Western Hemisphere, Jeffrey Wadsworth, lay relaxed under a cosmic-ray lamp, with no covering but a towel over his loins.

The surgeon-general of the Hemispheric Armies raised his hand, and the lamp receded.

"Is that enough?" Hoshawk asked dryly.

"It's the maximum, even for him," said the surgeon-general. "His reflexes will be faster than light itself."

Hoshawk grunted, his eyes narrow. As far as he could see, the speed of a man's reflexes, even of a man who was about to champion seven hundred million persons, wasn't as important as the man's loyalty or his sense of personal responsibility. And Hoshawk did not have much use for Wadsworth.

Augusto Iraola of Brazil, deputy president for South America, stepped forward from the group of forty men. He asked the President anxiously, "How do you feel?" Iraola was old and bearded.

"Not bad," said the President, and his voice squeaked a little as it changed pitch.

The Minister of State, with a big portfolio under his arm, said, "Shouldn't we prepare the vice president?"

Morrison, vice president for Canada, spoke pedantically, "It would be a tragedy to lose President Wadsworth. Last month his I.Q. was 340, nearly twenty points above any other member of the Mutant College."

Hoshawk barely caught himself in time to repress a snort. A boy of sixteen, no matter what his I.Q., was just a kid. You couldn't expect him to exhibit initiative or even to take things seriously. That was why Hoshawk had almost broken with the Hemispheric Congress thirty years before—almost two of President Jeffrey's lifetimes, Hoshawk reflected wryly.

The voice of the President, slightly amused, came to them. "I'm all right now," he said. "I think I ate too much ice cream last night. Nine dishes."

There were gasps. Hoshawk held back his sarcasm, but he could not refrain from a triumphant glance at the ancient Minister of State, who avoided his eyes.

Iraola was volatile. "Sabotage!" he said.

President Wadsworth licked his lips with the tip of his tongue. "No, the new pineapple-avocado. Very good, gentlemen. I recommend it."

The neuro-analyst whipped a graph from his machine. Hoshawk barely looked at the graph. "Speed of reaction down to zero, point, nine zeros, three, four—three times normal speed. Let's get on with the war."

The President's eyes had been fixed hopefully on Hoshawk's grizzled face, and at Hoshawk's words he relaxed. His muscles rippled an instant, and then he was standing.

It was always a little shock to Hoshawk to see him move. It wasn't right that any man, even a Superior Mutant, should be able to move faster than light-speed. You didn't dare to trust a man like that.

Forty august heads—all but Hoshawk's—inclined as the President stood there, but the President just smiled at them and yawned and stretched luxuriously.

Hoshawk was annoyed, but there was nothing he could do about it. The Hemispheric Congress had set up the Mutant College two hundred years ago, and every child with I.Q. above 200 and physique to match, became a member, for the sole purpose of selecting a President whose primary duty would be to fight a war, if it should come in his term, on one of the giant keyboards. This had been a concession to left-wing agitation that, if there was to be another war, it should be fought by the leaders and not by the ranks.

The Mutant College had been established when the Hunyas had overrun Europe and Asia, and now for two centuries there had been no war, but only preparation for war, East against West, through systems of selection and training closely parallel, but with a difference that was forever in Hoshawk's mind—if he was a capable man, the Hunyas kept him for twenty-one years.

And obviously you could depend a lot more on a man of thirty-five than you could on a boy of sixteen.

Forgacs, president of the Hunyas, was thirty-three—an old man for a mutant, and smart and clever as only a mutant could be at that age.

Yesterday the Hunyas had challenged.

It was sudden, but not unexpected. There was no reason for delay. At six o'clock tonight the two hemispheres would match force, and by eight o'clock it would be over.

Jeffrey Wadsworth moved. One instant he was before them with a towel around the middle of his bronze body, the next instant he was standing there dressed in light plastic slippers, red trunks and a sleeveless blue shirt. If Hoshawk hadn't been so old, he would have been envious of the President's physique.

"Gentlemen," Jeffrey said, "I am ready to go to the Chamber." He rubbed his bare midriff in the region of his stomach.

"Are you ill?" Hoshawk asked quickly.

"No," Jeffrey watched the forty statesmen file out.

"Sire," said Hoshawk, and his manner was respectful, for this boy of sixteen was his commander-in-chief, "I still wish we had trained a few thousand men in the use of weapons. I don't see how we can fight a war with electronic tubes."

Jeffrey looked at him gravely. "War with men is primitive. Lives can't be replaced."

Hoshawk sputtered. "There's never been any civilized war."

"This time there will be," Jeffrey said confidently.

"But—"

"We'll win," Jeffrey repeated. "We *must* win." And Hoshawk caught a flash of something deep in his eyes. Hoshawk could not quite identify it, and yet

he knew it spoke of the inner wisdom and conviction of the young. And in that direction, Hoshawk reasoned, lay their weakness.

"There'll be trickery from Forgacs," Hoshawk predicted.

"Quite possible," said Wadsworth. "I don't trust him, myself. He challenged on a technicality."

Hoshawk was gratified to hear a worried note in the President's voice. "He claimed we violated the Agreement of 2118," he said, probing, "by keeping scientific discoveries to ourselves."

Wadsworth answered quietly, "Then he challenged because he himself had secrets that he believed more potent."

"Nevertheless," said Hoshawk, "a few hundred men trained in the use of tanks—"

Jeffrey shook his head. "And revert to the primitive," he pointed out. "If the world is ever to get away from that kind of war, this is the time to prove it."

"And if we lose, we do so at the expense of a hemisphere."

"That's true," Jeffrey said calmly. "But if we should win by using men and destroying lives, we would do so at the expense of a civilization. By the act of reverting to the use of human fighters, we would convince the world that war could not be fought electronically."

They reached the door of the Chamber. The President shook hands with Iraola and with Hoshawk.

"Wish me luck," he said lightly.

They inclined their heads, and when they looked up, the President was seated on a beryllium stool that traveled a three-quarter circle before the great bank of keys like the keyboard of a giant organ. He pulled on a glass helmet and adjusted the sonic amplifiers to his mastoids. He flicked the oxygen valve open and shut, and then looked at it and listened intently.

Hoshawk saw an instant's doubt on the President's face. Hoshawk wondered if the valve was leaking, and frowned. The Chamber had been tested exhaustively, but with hundreds of thousands of circuits, cut-backs, by-passes, and relays, it was possible the oxygen valve had been overlooked.

Jeffrey strapped himself into the chair. The chronometer showed five minutes before the Hour. The President looked at the huge curved map of the Atlantic, now aglow with light above the big keyboard. His eyes swept the thousands of ivory keys and he rubbed his hands together for a final limbering of his fingers.

He spoke, and his intent voice came to them through the amplifier: "HHQ."

"North America is completely evacuated, Sire, to the Polar ice-cap. There is now no human being on the continent. The Hunyas refused our request to declare New York an open city, and it was evacuated thirty minutes ago."

The President called for a chronometer check. The instrument in the Chamber had lost two hundredths of a second, and Hoshawk could see that Jeffrey was making a mental note of that. He was forced to admit that the young mutant was thorough.

There were two minutes left. Jeffrey sat straight before the great keyboard, poised an instant, and then his incredibly facile fingers played the keys, flashing from one bank to the next, shooting the chair to right and to left, while he watched the map above him and the great bank of lights on each side. Then he leaned back, relaxed.

Hoshawk was glad now they were playing it safe. Jeffrey had insisted on the Midwest Chamber in preference to the Pacific or Atlantic station. For this was modern war. There would be only one person killed. This was a war of electronics, deadly and final, but no one would be actually killed but the losing President. That was decreed by the Six-Continent Council.

It was one minute before the hour. The President pressed a key.

The Starter answered: "President Wadsworth, are you ready?"

"Ready," said Jeffrey in a high voice.

Hoshawk heard the Starter's voice: "President Forgacs, are you ready?"

"Ja," came the deep voice of the Hunyas president.

Jeffrey flicked the oxygen valve for a second, snapped it off, and Hoshawk saw him glance down at it. Then Jeffrey sat poised, all the alertness of his incredible mind bearing intently on the map before him.

A bell sounded. The war was on!

Jeffrey did not move. He waited, and watched. Ten trillion electronic tubes would flash their information on the Map. He waited—one minute, two minutes, five minutes. The Map was dark.

So Forgacs wanted him to move first.

Jeffrey flicked the oxygen and his chair shot to the left. His fingers blurred into movement. He shot back to the center of the keyboard and focused his entire intellect on the Map.

A dozen tiny red lights rose off the coast of Newfoundland and raced eastward. Each light represented a thousand rockets loaded with thirty tons of DTN. One of those rockets would wipe Berlin from the earth—if it struck.

But Hoshawk knew the President did not expect them to reach Europe.

They did not. Near the coast of Holland they began to wink out. One got as far as Cologne.

If the Chamber had been above ground instead of three hundred feet deep in solid rock, they would have felt the concussion, for DTN's powerful waves traveled at the speed of light.

Still there was no answer.

Jeffrey's fingers played for an instant on the keys. Red lights rose from Labrador, from near Boston, from Florida, and streaked east—not for Berlin this time, but for Marseilles.

Jeffrey was testing Forgacs' explosive screen. It was wholly effective; one after the other, the trains of red lights winked out.

But now there was an answer. From the Bay of Biscay red lights with black dots on them began to wink on as the mammoth tabulating machine in the room below recorded the information from thousands of hidden electronic tubes, totaled it, and presented it on the Map.



The President hardly watched them. His screen with its principal power-plant in Philadelphia would stop the rockets, up to a total of some seventy-five octillion macro-ergs.

On the off chance that Forgacs would forget to close his screen after his rockets had passed it, Jeffrey fired a salvo from the Bahamas.

Forgacs answered with three salvos from Brest, and Jeffrey gave him back ten from Long Island, then Hoshawk frowned as he saw the President rub his stomach. Hoshawk had always opposed that abominable atavistic confection called ice cream.

It was a game of incredibly swift calculation and rapier thrusts from strong point to strong point in the effort to break through the screen. Once the

screen should be broken, anything might happen.

Jeffrey could see when his own screen was up, but their science had devised no way to detect the enemy's screen except by firing into it. Jeffrey pressed a pedal with his left foot, and a thin golden line flashed on in a flattened arc from Greenland down through the Atlantic and curved around the Falkland Islands.

Jeffrey's screen was up. The Biscay salvos began to wink out against it. Jeffrey's hands began to flash. Red lights winking up along the coast of Europe and from North Africa showed that Forgacs was opening up.

Jeffrey cut in the oxygen for a second and flicked it off, then his left foot slashed at the pedal as he cut his screen to let his own rockets through and then threw it on again to stop the enemy.

Forgacs was beginning a drive on Philadelphia, the site of the power plant. Jeffrey was watching for an opening to Marseilles, vulnerable for the same reason.

Jeffrey kept firing rockets, but his mutant mind would be racing ahead, calculating with infinite precision the times of discharge and times of arrival.

It was apparent by now that Forgacs' most powerful defenses were centered around Marseilles, because Forgacs was not using them. This meant he was not taking a chance on opening the Marseilles sector of the screen.

Jeffrey calculated the probable interchange of batteries for some sixty moves ahead, Hoshawk knew, then he began to fire the Philadelphia batteries at intervals.

The firing rose in intensity, and Jeffrey's faster-than-light fingers played the great keyboard like a master organ. A bell sounded and his right foot threw on the western screen with its automatic cut-out.

And all the time Jeffrey fired his big Philadelphia batteries at intervals with a definite rhythm—five, three, and six seconds.

He shot to the right and manipulated a bank of keys and was back in the center almost instantaneously.

He did not pause in his rocket salvos, but in three minutes and eight seconds his first salvo of one-ton atom bombs would reach the Marseilles screen. If he had calculated correctly, the Marseilles screen would be open for an instant just as the atom bombs reached it. He didn't think Forgacs could resist the temptation to blast Philadelphia with his Marseilles batteries.

Presently a thousand red lights winked up from the screen at Marseilles. But Forgacs overlooked the atom bombs. They were slower than the rockets, and there was no way to tell, from the Map, which was which.

Jeffrey shot a look at the chronometer, and Hoshawk saw the atom bombs go through. A few seconds later the glow in Marseilles began to redden, and Hoshawk exulted. The atom bombs had done their work. The Marseilles screen was weakening.

Jeffrey played the keys with fantastic speed. The war would soon be over. Thousands of little red lights began streaking toward Marseilles. At first they exploded in air as they hit the screen, but as the explosive force of the DTN began to drain the screen, those behind began to pour through.

But there was a flash from Philadelphia, and a shock went through Hoshawk. Something was wrong there. Jeffrey hadn't intended that. Forgacs had used atom bombs and had broken through when the screen was down.

Jeffrey's fingers snatched at the oxygen valve. He tore it off and threw it on the floor. He still held one important advantage. He was ahead of Forgacs by forty seconds.

Philadelphia went out and the golden defensive screen began to fade, but Jeffrey, tensely erect, stayed on the attack. Hundreds of green lights began to rise around Marseilles—great submarines, controlled by electronics and carrying tanks and guns and explosives.

The green lights converged on Marseilles. They got through the screen. Now was the big gamble. Jeffrey guessed that Forgacs would operate from an underground chamber near Marseilles itself.

It wasn't a logical thing to do, and so Forgacs would do it, believing that Jeffrey would pass Marseilles and go inland to find the Chamber.

Jeffrey let him believe that. He sent eight thousand giant electron-controlled bombers through the Marseilles gap and straight for Berlin.

The green lights started winking on the coast of France, showing the submarines were unloading amphibious tanks. Jeffrey started them out across France at high speed. Near Paris they met heavy resistance from Forgacs' tank-killers.

But now Jeffrey had more trouble. Forgacs had slipped a salvo of atom bombs into the Labrador power station, and the entire north quadrant of Jeffrey's screen was down. And just at that instant, the automatic breaker failed and a tube burned out in the Montevideo power station, and the southern half of South America was exposed. Green lights began to wink up at the open spaces.

Jeffrey was grim. It was near the end. Dog eat dog. His flying fingers chose to ignore Forgacs' attack, beyond firing millions of salvos of small rockets which were little better than a delaying action.

There were only two targets in this war—the Chambers.

Jeffrey released his trump—thirty-five hundred flying robot tanks.

They rocketed through the Marseilles screen and came on the city from the land side, firing eight-inch rockets and shooting flames out half a mile ahead.

But this was a feint, too. From the sea now rose a great armada of robot submarine carriers that spewed out tanks that were little more than armored tank-cars filled with jellied XPR, which exploded always down, toward the center of gravitation. They poured out the jelly on the surface around Marseilles for a distance of twenty miles until according to Jeffrey's figures the ground was covered a foot thick. The flame-throwers roared into it and Jeffrey stopped them there.

Then he fired his last salvo of atom-bombs from the Bahamas.

In the meantime, Forgacs' tanks had overrun Boston, searching for the American Chamber.

The lights began to wink out, and Hoshawk knew that Boston was being destroyed.

Orange lights, indicating bombers, were heading for Chicago, and Hoshawk knew that if Jeffrey's guess on Marseilles was bad, he had not much longer to live.

He looked at the Map. The atom-bombs were at Marseilles. A glow showed around the twenty-mile circle that he had covered with jelly, and Hoshawk knew the atom-bombs had landed.

He knew that on the other continent, the most tremendous explosion in man's history was taking place. And when it was over there would be a mile-deep crater where Marseilles had been, and anything, no matter how deep it was buried, would be destroyed by concussion.

Jeffrey still played the keys, but his eyes were on the orange lights approaching Chicago.

They reached Chicago, perhaps directly over their heads, but Hoshawk felt no bombs. A moment later the planes were still going westward.

Jeffrey called the Starter. "Does Forgacs concede?" he asked.

There was a moment's delay, then, "Forgacs does not answer."

The President let out an undignified whoop. He tore off the straps that held him in the chair, threw his helmet across the Chamber. "We won!"

The Hemispheric diplomats were gathering excitedly in the corridor. Jeffrey unsealed the Chamber.

Hoshawk shook hands with him. "You did it," he said gruffly. "I apologize for ever thinking—"

The Chamber shuddered, and Hoshawk paled, but Jeffrey held up his hand. He glanced at the chronometer. "That was Marseilles blowing up," he said.

His feet moved and he was gone. In a moment he was back. "Excuse me, gentlemen," he begged. "I've got to see the squad. Just figured out a way to beat the Blues. If you—"

He stopped, frowned.

He had felt it before they did—a distant blast. Then they heard it—a dull explosion through three hundred feet of solid rock above them. The floor shuddered under their feet.

It came again, and again, farther away. A pattern. Then off somewhere else came another string of explosions.

The forty august heads stared at the ceiling. Mouths were open, but the President's mutant brain in seconds analyzed the possibilities and came up with the answer:

"Atom-bombs!"

"Impossible!" growled Hoshawk. "Forgacs' Chamber was destroyed."

The President was already back in the Chamber. He pressed a key.

"Starter," came the answer. "Forgacs' Chamber is destroyed. You have won the war."

Hoshawk was behind him. "But he's still firing, isn't he?"

"No." The President was icily alert. He pointed to the big map. There were no red pin-points that would indicate rockets or bombs coming from the European continent. "The Chamber is gone. Undeniably gone."

A new pattern of bomb-bursts came from above. "Chicago must be destroyed by now," said the President harshly. He pointed to a blacked-out area on the ground-glass screen above. "There are no detector tubes left above us. But look—orange lights. Thousands of them coming from the sea on the Maryland coast. And look there, to the right. One—two—fifteen thousand bombers coming!"

Hoshawk nodded as if he had known it all the time. "Sure. He has men in those planes. Live men who can observe and act independently. He's throwing hundreds of thousands of planes and submarine tractors and mobile bomb-throwers at us—all operated by men. And Forgacs himself is here, leading them. We're whipped, Sire! Where is your civilization now?"

Wadsworth was calm. He was taking it like a man, anyway. He threw a lever and poised at the great keyboard, then his mutant fingers began to work in blurred movement.

Hoshawk watched the screen above. The Atlantic filled with long trains of red lights that arose from their American bases and streamed eastward.

Hoshawk blinked. "You're firing everything. And you've locked the controls."

Wadsworth didn't look up. "In five minutes," he said, "there won't be an ounce of explosive left in any emplacement in America."

"But that's—" Hoshawk started to say "foolish," but he changed it. "That won't help, Sire. Forgacs' equipment is all over here, now."

But Wadsworth leaned back. Their golden explosive screen showed no longer on the Map. Already some of the emplacements had ceased to spew out red lights, and the tail-ends of their trains were disappearing to the east.

Hoshawk shuddered as he saw that now America was completely defenseless.

But Wadsworth spoke into his transmitter. "Radio. Give me special frequency three-hundred-eighty-one thousand, six hundred kilocycles. Clear all air-lines."

"Yes, sire."

The President pressed the scrambler button and then spoke. The words came out of the amplifier. "Three tons of butter unloaded a fast curve day before tomorrow because the baby was yelling for its morning high-ball. The soap-suds are thick enough for whipping but who knows where or when."

The President leaned back and smiled. "That's an order to all sixteen thousand mutants over the country to be on the alert at their predetermined stations."

Hoshawk frowned. "But everybody's been evacuated."

"Not the mutants. You see General, we ourselves haven't trusted Forgacs."

Hoshawk's grim face lighted up. "Do you mean you have secretly made some fighting equipment?"

Wadsworth shook his head. "No. We could have. There's a loophole in the Twenty-one Eighteen Agreement. But we have observed the spirit—ah!"

Up on the ground-glass screen, purple lights had been flashing on at intervals over the United States, until now there were nineteen, and Wadsworth spoke: "Those represent transmitter stations equally spaced over the country. They are all manned by mutants."

Hoshawk actually snorted. "Transmitter stations! You can't fight with words! And, anyway, there won't be any power at all within a half hour."

"They each have their own power-plant," The President said quietly.

Hoshawk looked at the map again and groaned. The nation was almost covered by a canopy of orange lights marked with black crosses. "There must be at least a million bombers over us! They'll wipe out the whole country within an hour. If there's anything you can do, *do it!*"

The President was pale, but he sat quietly. "Stalled," Hoshawk thought sardonically. It took something besides smartness to win a war. It took character, too.

Wadsworth pointed to the American shores. Long lines of green and white and black and yellow dots coming from the sea, crawling in among the orange lights that swarmed over America like a gigantic swarm of hornets. "Submarines, amphibian battleships, flame-throwers, tanks," he said.

Hoshawk stood erect. "If it were not against regulations, Sire, I would be tempted to blow my head off. We shall be destroyed as a people and as a continent."

The President's hands were clenched, but he answered slowly, "As a continent, perhaps. But the buildings can be built again. As a people—no, I don't think so. As a civilization, I hope we can be saved."

Hoshawk's eyes narrowed. "How?" he demanded.

"Those purple lights represent sonic transmitters. In other words, generating stations for sound frequencies above the narrow band which can be heard by humans. They were developed, built, and financed by graduate mutants. They broadcast on different frequencies that we have determined most effective in upsetting the equilibrium of unstable chemical compounds."

"Do you mean," asked Hoshawk, "that you are going to try to detonate the explosives carried by Forgacs' planes?"

"His planes, and anything else that carries them. We have analyzed samples of his explosives to determine the critical frequency of each. These nineteen stations cover the country. Any known explosive in the continental United States will be detonated when these stations go into operation."

"What if Forgacs has some unknown explosive?"

Wadsworth was solemn. "We take that chance," he said. "But the range of possible explosive combinations is well known, and something entirely different is unlikely. At any rate—"

"They're starting to drop bombs!" Hoshawk said.

The President watched the red glow around Kansas City. His face was taut. "There will be many cities destroyed," he said. "But we must wait for all of Forgacs' equipment to be within our continental limits. It must all be destroyed at once."

"But the bombers are in action," said Hoshawk. "Denver is getting it now."

Wadsworth's eyes were on the coastlines. "It will be twenty minutes at least before we can open the transmitters. We may lose most of our cities by that time, but there is nothing we can do."

The red glows began to spread. Dallas and Fort Worth, New Orleans, Atlanta, Miami, San Diego and Los Angeles, San Francisco, Portland and Seattle. The bombers were systematically destroying America's population centers. And still Wadsworth waited. He sat tense before the Map, watching the endless stream of lights come from the sea.

But they were beginning to end. Many were far inland, attacking the smaller cities, cleaning up the big ones.

"The bombers won't be destroyed," said Hoshawk, "if they've already dropped their bombs."

"I think they will, for all practical purposes," said the President. "Their ammunition, their signal flares—everything explosive will be detonated."

"How can you cover them all at once?"

"There are over nine hundred frequencies—but we don't know that they will be enough," Jeffrey pointed out gravely. "We can only hope."

Hoshawk couldn't stand still any longer. He paced the floor before the Map. "Every city in America of more than a hundred thousand is gone—obliterated," he said tonelessly. "Can't we ever—"

"Wait!" The President was alert. "The last line of flame-throwers is coming on land." He pointed to the black dots streaming up on the west coast. He spoke into the audio transmitter. He didn't bother with the scrambler now. "Sonic stations on. Emergency force. Sonic stations on. Emergency force. Situation critical."

He pointed to the Map and sat back. Within a few seconds the purple lights began to flash intermittently.

"They're on," said the President. "But it will take a few minutes for them to reach full intensity. The sonic devices operate at high speeds—some at two hundred thousand r.p.m."

Hoshawk watched, almost without breathing. For the first time he was aware that the forty statesmen of the Western Hemisphere were watching through the glass windows of the Chamber.

At that instant purple glows began to surround the green lights, starting on the east coast of Florida and spreading upward.

"Amphibian submarines," whispered the President. "Their aerial torpedoes are exploding!"

"And up around the Great Lakes," said Hoshawk. "There it's amphibian tanks."

The President sat, and watched. The glows spread. They absorbed flame-throwers, tractors, mine-heavers. The Map of America was a clustered mass of lights, with the purple glow beginning to consume everything in its reach.

"The planes," said Jeffrey. "They're still untouched. They anticipated something like this." He barked into the microphone. "All stations, ascending frequency!" he ordered, and turned to Hoshawk. "We don't know how effective this will be. It isn't as powerful as the static ranges. But—"

"It is! They've got the range!" cried Hoshawk.

Jeffrey looked. Near Albuquerque, New Mexico, a cluster of orange lights was being consumed by the purple glow. Jeffrey shot a glance at a dial. "All stations! All stations! Frequency seventy-two thousand, nine eighty. Emergency. Frequency seventy-two thousand, nine eighty."

And the purple glow rolled and spread and consumed Forgacs' bombers by the thousands.

At last Wadsworth looked at the Map, with nothing left but the dead embers of a mighty army.

Hoshawk shook hands with him and then looked for a place to sit down for a moment. "Sire," he said at last, licking his lips with the tip of his tongue, "if it isn't presumptuous, I'd stand the check for a dish of that new ice cream."

Jeffrey looked at him and smiled. "You'd better have one yourself."

Hoshawk's grizzled face was solemn. "I'm going to," he said.



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